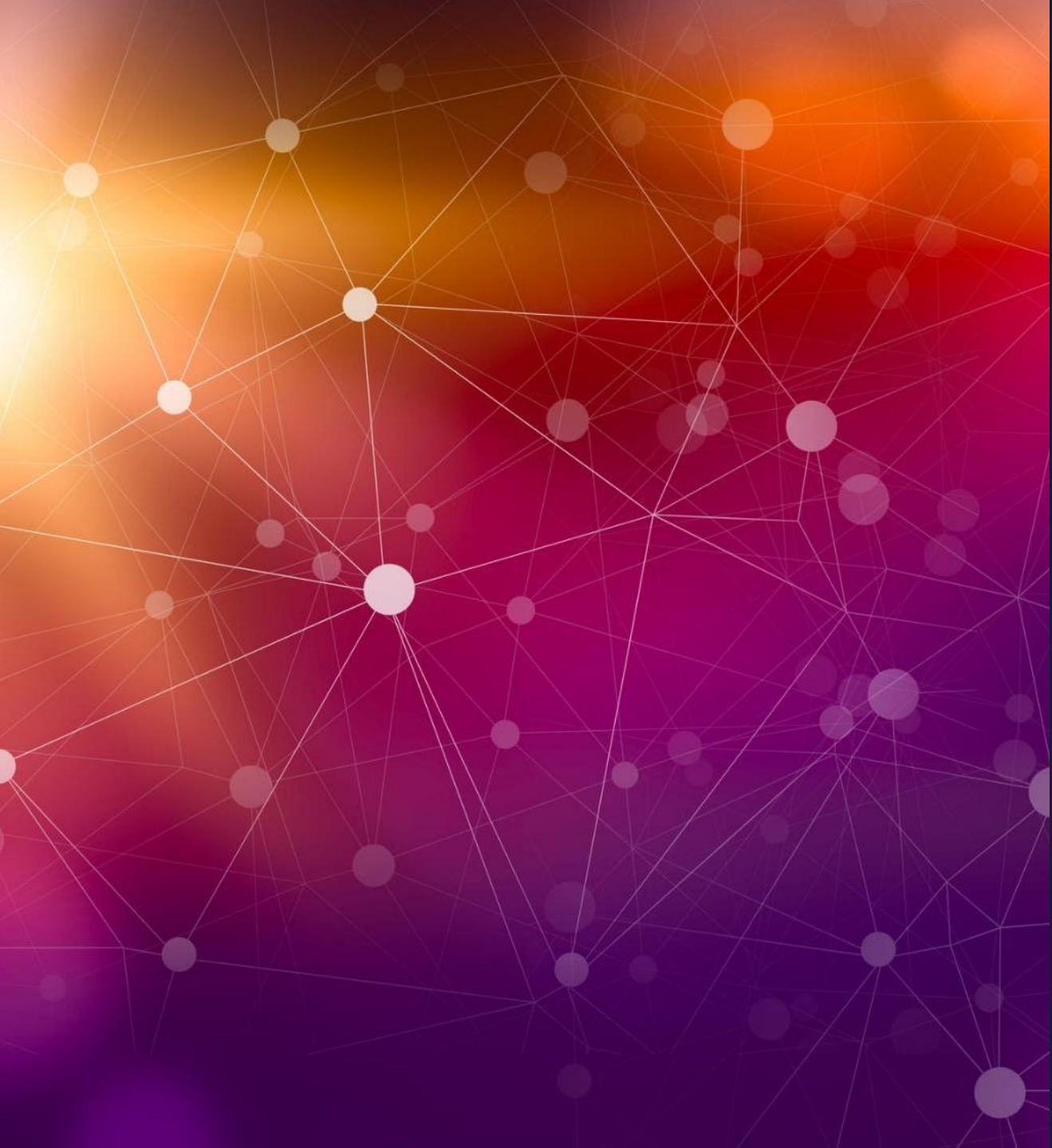




Analytics in the Octagon: Modeling UFC Fight Prediction

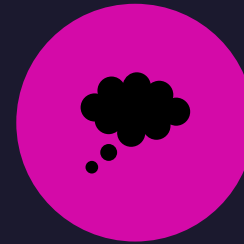
Group 4 -
Evan Embry

CSPB 4502, University of
Colorado-Boulder

Questions Sought to Answer



CAN FIGHTER
ATTRIBUTES
AND DERIVED
METRICS
PREDICT THE
OUTCOME OF
A UFC BOUT?

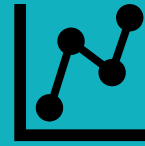


HOW DO OUR
PREDICTIONS
COMPARE WITH
BOOKMAKER
ODDS?



WHICH
MODELING
APPROACH
(LOGISTIC
REGRESSION,
RANDOM FOREST,
XGBOOST)
PERFORMS BEST?

Data Preparation Work



Data Set: “Ultimate UFC Dataset” from Kaggle



Cleaning Steps: Dropping irrelevant columns, imputing missing values, label encoding

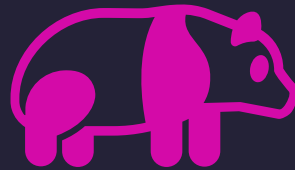


Feature Engineering:
Created ReachDiff, HeightDiff, AgeDiff

Tools and Technologies



**Programming
Language:** Python



Libraries: pandas, numpy,
scikit-learn, xgboost



Environment: Jupyter
Notebook

Modeling Techniques



Logistic Regression (baseline)



Random Forest (ensemble method with hyperparameter tuning)



XGBoost (tuned with GridSearchCV for optimal performance)

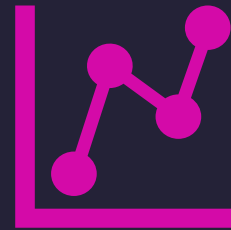
Key Findings & Knowledge Gained



Results:

Random Forest achieved ~64.5% accuracy and a 70.5% F1 score

Tuning XGBoost improved its performance to be competitive



Insights:

Derived features (reach, height, age differences) are crucial predictors

Ensemble methods capture nonlinear dynamics in fight data



Practical Applications

Coaching/
Training

Coaching/Training: Use insights to tailor fighter training programs

Betting

Betting Analytics: Identify discrepancies between model predictions and bookmaker odds

Broadcasti
ng

Broadcasting: Enhance live commentary with real-time analytics

Thank you

